DATS 4001
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1. **Learning outcome**
   While I did not begin with a specific research question. My initial goal was just to build a website that solved a real-world social media cyber problem I'd actually experienced. As I worked, I stopped looking for a broad solution and focused on the technical mechanics of user experience. I learned that a project does not need a formal hypothesis to be successful. Solving a personal frustration taught me more about practical web development than following a rigid proposal ever could.

2. **Challenges and problem-solving**
   Coding the models, data processing, and web interface was my biggest challenge. I handled this by researching specific codes on Youtube and Google to output the expectation I want, then adapting that logic to fit my project's architecture. This required a lot of trial and error to ensure the data cleaning and web design actually worked together. In addition, when I hit a wall with error messages I could not interpret, I use ChatGPT to break down the feedback and pinpoint the bug. It helped me understand what the machine was actually complaining about. On the version control side, I struggled with GitHub not updating my files correctly. I learned to be more disciplined with my workflow, double-checking my commits and pushes to ensure the latest version of the code appeared on the GitHub platform.

3. **Ethics**
   An ethical consideration in this project involves the privacy concerns of users regarding their digital identities. While the research only uses public information like follower counts and following lists, people often feel uneasy when their data is taken for studies without their consent. Even public data can be misused if it is used to track or profile individuals in ways they did not consent to on the platform. To address this, I focused strictly on aggregated metrics rather than individual identities. I ensured that no personally identifiable information was stored or highlighted in the final web output. To further improve the ethical standing of the project, I could have implemented a data masking technique to ensure that specific users cannot be traced back from the research results. This protects the privacy of the participants while still allowing the project to analyze broader social trends.

4. **Use of generative AI**

   I used Gemini AI to generate the background images for my website's form, result, and error pages. I verified I understood the output by judging the background picture generated to match the project I'm focusing on. I also used ChatGPT to debug my code

and find specific errors that were stopping my web from running or my model/plot from outputting the correct analysis. It helps me clean my data and figure out the best way to structure my visualization plots. I verified this work by manually fixing the identified bugs and making sure to test that the code ran perfectly without any errors.

5. **Reflection on your process**

   If I were starting this project again, I would change how I integrate machine learning into the web platform. Instead of manually writing code based on observed behaviors. I will focus on learning advanced implementation techniques to pipe the models directly into the site. This would make the detection process more robust and professional. With more time extension, I would expand the feature set. I will research and include more effective indicators of impersonation to drive higher accuracy and provide more definitive results for the end user. Finally, I would slow down the initial planning phase. I need to better estimate how long tasks like data collection and result verification actually take, so the end of the project doesn't feel rushed. Moving forward, I'll use literature reviews and similar research papers as a benchmark to more accurately predict my own timeline for each stage of the process.

6. **Anything else**

   I think learning technical tools like the GitHub command line and HTML in the prerequisite class before starting the capstone class reduces my overwhelm. Learning them for the first time while trying to build the project was overwhelming and took way too much time away from the actual research. Once I finally understood the tools and had hands-on practice beforehand. I had much more time to focus on the project itself. Having that technical foundation earlier would let us use the capstone class time more effectively for the project.